



Development of AI model for gender differences in bicycle mobility of green urban areas for recreational and educational institutions of cycling behaviour, patterns and constraints of regular cyclists for Chandigarh city—an Indian perspective

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Abstract

Gender differences in cycling patterns, motivators, and constraints in Chandigarh, India, were studied by a team of researchers from Punjab Engineering College. The primary objective of this study is to explore the gender-specific differences in cycling behaviour, including the distinct patterns, motivators, and constraints experienced by male and female cyclists. A secondary objective is to provide actionable policy recommendations to enhance cycling infrastructure and promote cycling as a sustainable mode of transportation and recreation in Chandigarh. The researchers collected data on traffic flow at intersections in different zones, conducted online and field surveys, and analysed regulations and guidelines regarding Non-Motorized Transportation modes established by local governmental entities in Chandigarh. The study found that male cyclists predominantly used bicycles for commuting short distances efficiently, while females gravitated toward recreational cycling. Safety concerns were identified as a deterrent to daily commuting by bicycle, particularly affecting female cyclists. Dedicated cycling infrastructure and gender-sensitive policies were found to be vital for increasing female participation in cycling. Social norms and responsibilities also influenced women's cycling habits. Despite gender differences, both genders shared an interest in cycling for exercise and recreation. The study emphasized the need for gender-sensitive policies, urban planning initiatives, and improved safety measures to promote cycling as a sustainable and safe mode of transportation and recreation in Chandigarh. Improving safety protocols for cyclists of all genders is vital to reducing accidents and promoting cycling as a safe and practical mode of transportation within the city. The study provides evidence for making effective policy changes and infrastructure improvements to make Chandigarh more inclusive and biker-friendly. Promoting cycling as a sustainable mode of transportation and recreation among women is crucial for developing inclusive and sustainable urban areas. The researchers proposed machine learning techniques such as decision trees, random forest, and gradient boosting to examine the relationship between gender and cycling patterns, motivators, and constraints. Thematic Analysis was used on collected qualitative data to identify recurring themes, patterns, and nuances. Analysis of the current cycle tracks in Chandigarh was done using photographic surveys and public polls. The team identified a lack of data on gender differences in cycling in Chandigarh, India. Such data is essential to improve the conditions of cycle tracks in the city since men and women have different cycling requirements. The study highlights the importance of gender-sensitive policies and urban planning initiatives in Chandigarh. In conclusion, the study provides valuable insights into gender differences in cycling patterns, motivators, and constraints in Chandigarh, India. The study emphasizes the need for gender-sensitive policies, urban planning initiatives, and improved safety measures to promote cycling as a sustainable and safe mode of transportation and recreation within the city.

Keywords Chandigarh administration · Cyclist fatalities · Recreational cycling · Cycle tracks · Gender differences · Mode transportation · Safety

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Literature review

Chandigarh, renowned for its urban planning and modern infrastructure, has witnessed a surge in cycling as a mode of transportation and recreation [1]. However, the existing knowledge base regarding gender-specific patterns in cycling within the city remains inadequate. This lack of localized data hampers the effectiveness of ongoing initiatives aimed at fostering favourable conditions for cyclists, particularly when considering the distinctive requirements and preferences of individuals based on gender [2]. Cycling provides health benefits such as enhanced cardio-respiratory fitness and a reduced risk of overall and cardiovascular mortality [3]. Previous analyses found that driving to work was significantly associated with being overweight and obese [4]. Grounded in a social constructionist perspective of gender, the anticipation is that cycling behaviour will diverge based on socially constructed notions of masculinity and femininity [5]. The primary aim of this research is to conduct a comprehensive investigation into the gender-specific differences in cycling behaviour within Chandigarh. The research will examine participation rates and delve into the finer details of recreational and commuter cycling. By pinpointing the specific patterns, motivators, and constraints that impact men and women differently in their cycling activities [6], this study seeks to provide valuable insights for urban planning and policy development [7]. This paper aims to target the worker class and identify the problems faced by them during a cycle trip. The maximum percentage of trips by 'other workers' in urban India (more than one-third of trips) are in the 1–5 km range, where a cycle is regarded as the most efficient mode of transportation. Nearly 60% of all trips occur within a 5 km radius and approximately 80% within a 10 km radius [2]. Young white-collar workers working in office settings, residing in tall apartment buildings, and actively participating in recreational cycling with groups are promising demographics to target for promoting cycling as a means of commuting. On the flip side, an unexpected discovery was that, when compared to those who drive to work, individuals who use public transport are more inclined to transition into being regular commuter-cyclists [8]. In addition to analysing individual cycling behaviour [9], this research will also conduct a thorough review of the rules and policies governing Non-Motorized Transportation (NMT) [10] modes in Chandigarh. By assessing the current regulatory landscape, the study aims to identify any potential gaps and areas for improvement in promoting cycling as a sustainable and inclusive mode of transportation [11]. Through this multifaceted approach, the study aims to provide a comprehensive understanding of the gender dynamics of

cycling in Chandigarh. Previous research has significantly informed the methodology and analysis of this study. Traffic flow analyses at intersections, guided by earlier studies on cyclist behaviour and safety, were conducted to understand peak cycling times and safety issues. Online and field surveys, inspired by common methods in prior research, were utilized to gather comprehensive data from a broad demographic, ensuring inclusivity and capturing diverse perspectives. The review of local NMT regulations analysed how these policies impact cycling infrastructure and safety, identifying gaps in current regulations. Safety concerns and infrastructure conditions were assessed using photographic surveys and public polls to provide targeted recommendations. Machine learning techniques, such as decision trees, random forests, and gradient boosting, used in transportation studies, were employed to examine the relationship between gender and cycling patterns. Thematic analysis of qualitative data from surveys and interviews identified key themes related to gender differences in cycling, providing deeper insights into motivators and barriers. The study also analysed the impact of social norms and responsibilities on cycling habits among men and women in Chandigarh, examining cultural attitudes towards cycling. Ultimately, the goal is to offer evidence-based recommendations that can guide the improvement of cycling infrastructure and the development of gender-sensitive policies, resulting in a more inclusive and sustainable urban environment [12].

Research methodology

Sampling and study protocol

A comprehensive study was conducted to thoroughly evaluate the attitudes, behaviours, and cycling experiences of individuals of all ages who enjoy cycling. Participants were provided with both online surveys and field survey questionnaires to complete discreetly. The sampling was carefully segmented into three age groups (under 18), (between 18 and 65), and (over 65), and further categorized by assumed gender (Table 4). Moreover, meticulous data on bicycle traffic at intersections was collected by a team of five researchers utilizing click counters and smartphones (Tables 1, 2). This sample comprises a diverse group of individuals, including a portion of highly educated individuals who were surveyed in reputable institutions throughout Chandigarh. Most of the sample consists of working-class individuals who rely on bicycles for their daily commute. The objective is to determine if engaging in recreational activities influences the cycling habits of everyday individuals (Table 9) [6]. Collecting and analyzing data on cyclist fatalities and the conditions of cycle tracks is essential for improving

Table 1 Site selection

Intersection	Zone	Adjacent sectors/Areas
A	Educational	1, 11, 12, Nayagaon
B	Welfare	11, 12, 14, 15
C	Welfare / Educational	10, 11, 15, 16
D	Residential	2, 3, 10, 11
E	Educational	14, 15, 24, 25
F	Educational	15, 16, 23, 24
G	Residential	3, 4, 9, 10
H	Residential	4, 5, 8, 9

Table 2 Data on pedestrian count and LOS

Time interval	No. of pedestrian	Flow rate (Ped/min/m)	Space (m ² / Ped)	Level of service
1:00–1:15	54	0.108	55.55	A
1:15–1:30	51	0.102	58.82	A
1:30–1:45	55	0.011	545.4	A
1:45–2:00	57	0.0114	526.3	A

Table 3 Data of pedestrian count and LOS

Time interval	No. of pedestrian	Flow rate (Ped/min/m)	Space (m ² /Ped)	Level of service
1:00–1:15	117	0.0234	256.4	A
1:15–1:30	125	0.025	240	A
1:30–1:45	127	0.0254	236.2	A
1:45–2:00	121	0.0242	247.9	A

safety. This includes evaluating factors such as track surface quality, signage, and adherence to safety standards. Uneven or poorly maintained surfaces, inadequate signage, and non-compliance with safety protocols can all increase the risk of accidents. This data on cyclist deaths was collected from official records and Chandigarh administration sources. The team took a proactive approach by distributing email invitations containing a survey link to multiple institutions to encourage student participation. Additionally, the survey was shared through the personal connections of team members to ensure a diverse and extensive sample pool. Furthermore, the team actively sought survey responses from bicycle riders on the road, security guards, and BPL (Below Poverty Line) individuals (Table 3). A sample of 1687 bicyclists was collected during peak hours from various recreational, educational, residential, and industrial zones (Table 4). Additionally, a sample of 114 cyclists was collected to examine the health conditions of the cyclists,

as well as the motivators and constraints for commuter and recreational cycling (Tables 9, 10).

Site selection

Chandigarh is a well-planned city that offers a variety of cycling environments, with well-maintained roads, extensive cycle tracks, and a flat terrain ideal for cycling year-round. Its cycling infrastructure plays a vital role in determining cycling patterns, and the city has an extensive network of dedicated cycle tracks, promoting cycling as a viable mode of transportation. Safety is a priority, with designated bike lanes, traffic calming measures, and awareness campaigns, although challenges such as cycle-track encroachments and inadequate lighting still exist. Accessibility is facilitated by Chandigarh's well-connected layout, bike-sharing schemes, and bicycle rental services, catering to a wider demographic of cyclists. The city's relatively clean air and moderate climate make it favourable for cycling year-round, although seasonal variations may pose challenges that can be addressed by providing shaded rest areas and promoting cycling during optimal weather conditions. In summary, Chandigarh offers a diverse range of cycling environments, prioritizes cyclist safety, and has accessible cycling infrastructure, making it an ideal city for cycling enthusiasts to explore.

Site conditions

The data was collected at intersections during peak hours. The number of cyclists and other NMT modes has been counted separately on the cycle tracks and the main road. The footpath is covered with food and accessories stalls. The garbage generated is not maintained well. The footpath and cycle tracks are not well maintained. The dimensions of the 100 m section and the photograph accurately represent the location and its current condition as shown in Figs. 1 and 2.

Intersection traffic count

In urban and suburban areas, traffic jams at intersections can lead to a host of issues. Congestion can cause delays, long wait times, and frustration for drivers, while also increasing the risk of accidents. Safety is a top concern at intersections due to conflicting movements and the potential for distracted or hostile drivers. Efficiency is also impacted by factors such as traffic signal timing, lane design, and pedestrian and cyclist flow. While solutions such as optimal signal timing and advanced intersection designs can help, technological advancements like smart traffic management systems and self-driving cars may provide even greater improvements. Ultimately, addressing these congestion, safety, and efficiency challenges requires a multi-pronged

Table 4 Traffic at intersections

Intersection	Zone	Total	Age					
			< 18		18–65		> 65	
A	Educational	490	203		249		38	
			M	F	M	F	M	F
			166	17	195	35	19	1
B	Welfare	387	94		271		22	
			M	F	M	F	M	F
			86	8	224	47	22	0
C	Welfare/ Education	246	104		119		23	
			M	F	M	F	M	F
			94	10	94	25	23	0
D	Residential	148	11		124		13	
			M	F	M	F	M	F
			10	1	98	26	13	0
E	Educational	125	24		86		15	
			M	F	M	F	M	F
			23	1	74	12	15	0
F	Educational	130	41		84		5	
			M	F	M	F	M	F
			38	3	70	14	5	0
G	Residential	72	24		41		7	
			M	F	M	F	M	F
			22	2	30	11	7	0
H	Residential	89	18		68		3	
			M	F	M	F	M	F
			15	3	50	18	3	0

**Fig.** Intersection at PGIMER, Chandigarh—Depicting current conditions of footpaths with food and accessory stalls and inadequate garbage maintenance in front of Nehru Hospital (NH) Block**Fig. 2** Intersection at Vidya Path near Punjab Engineering College—Depicting the current condition of the location for the counting of cyclists and other Non-Motorized Transport (NMT) users

approach that includes engineering, technology, and policy efforts (Tables 5, 6).

This table provides a comprehensive analysis of traffic flow at various intersections, categorizing cyclists by age and gender. It offers valuable insights into the dynamics of cycling patterns, contributing to a nuanced understanding of intersection usage. The systematic classification and analysis of traffic flow aid in comprehending the impact

of different factors, such as location and demographics, on cycling patterns at various intersection sites [13]. The data gathered can be used by policymakers, urban planners, and researchers to customise interventions, infrastructure improvements, or safety measures according to cyclists' preferences in different urban environments (Table 7), leading to improved cycling experiences (Figs. 3, 4).

Table 5 Moderate MET IPAQ scoring

Characteristics		Moderate MET	Sample	% of total sample
Days of activity per week	Minutes of activity per day	n	n	%
Daily	15	420	78	4.335
	30	840	81	4.497
	60	1680	126	6.996
Weekly	15	60	142	7.884
	30	120	205	11.382
	60	240	95	5.274
Rarely	15	0	537	29.816
	30	0	300	16.657
	60	0	237	13.159

Cycling is considered a moderate MET activity

Table 6 Road crash data (2018–2023)

Road crash data								
Year	Total cases	Fatal cases	Killed persons		Pedestrian deaths		Cyclist deaths	
2018	316	97	98		35		11	
			M	F	M	F	M	F
			83	15	27	8	10	1
2019	305	100	104		36		10	
			M	F	M	F	M	F
			84	20	25	11	8	2
2020	159	50	53		12		10	
			M	F	M	F	M	F
			46	7	9	3	10	0
2021	208	94	96		24		17	
			M	F	M	F	M	F
			79	17	18	6	15	2
2022	237	79	83		23		12	
			M	F	M	F	M	F
			68	15	16	7	12	0
2023 up to 27.05.2023	68	24	27		8		6	
			M	F	M	F	M	F
			21	6	5	3	5	1

Research approach and AI model for predicting cyclists perceived safety

This research explores the predictive analysis of factors influencing cyclists' perceived safety while cycling. By employing machine learning models such as Decision Tree Regressor, Random Forest Regressor, and Gradient Boosting Regressor, the study identifies key predictors and evaluates the performance of these models in terms of Mean Squared Error (MSE) and R-squared (R^2) metrics. The results offer valuable insights for urban planners and policymakers to enhance cyclist safety.

Data collection and preprocessing

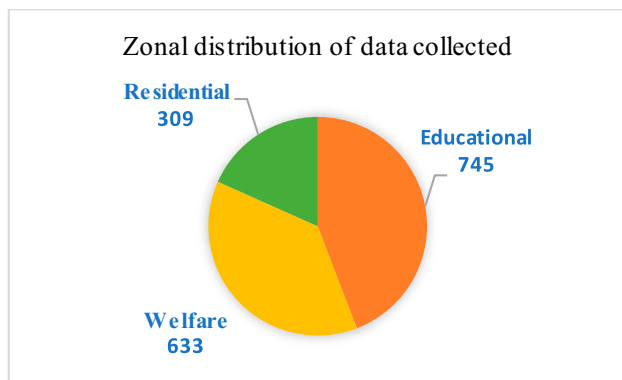
The dataset for this study was compiled from multiple sources, including:

- Official reports on road infrastructure conditions (e.g., faulty roads, manholes).
- Historical accident data from transportation authorities.
- Demographic data on cyclist victims from surveys and government databases.

Preprocessing Steps:

Table 7 Road Crash Data (2018–2023)

Road crash data					
Year	Two-wheeler deaths		Non-fatal cases	Injured persons	
2018	40		219	300	
	M	F		M	F
	35	5		232	68
2019	46		205	275	
	M	F			
	41	5		216	59
2020	20		109	148	
	M	F		M	F
	20	0		117	31
2021	32		114	172	
	M	F		M	F
	29	3		135	37
2022	40		158	203	
	M	F		M	F
	35	5		164	39
2023 up to 27.05.2023	9		44	67	
	M	F		M	F
	7	2		53	14

**Fig. 3** The pie chart displays the use of cycling at crossings for varied purposes

- **Data Cleansing:** Removed duplicates and handled missing values through imputation.
- **Standardization:** Standardized numerical features using normalization methods.
- **Encoding:** Categorical variables were encoded using one-hot encoding or label encoding.

Methodology

Data Cleaning:

- Removed rows with missing values.

- Excluded irrelevant features such as respondents' names and non-informative columns.

Model training and evaluation

- Split the data into training and testing sets.
- Trained three machine learning models: Decision Tree Regressor, Random Forest Regressor, and Gradient Boosting Regressor.
- Evaluated the models using Mean Squared Error (MSE) and R-squared (R^2) metrics.

Model results

The evaluation of the models yielded the following results:

- **Decision Tree Regressor:**
Mean Squared Error: 5.66666
 R^2 Score: -2.14383
- **Random Forest Regressor:**
Mean Squared Error: 1.4096
 R^2 Score: 0.21791
- **Gradient Boosting Regressor:**
Mean Squared Error: 1.2882
 R^2 Score: 0.28530

Implications of the model results

The performance metrics of the models suggest several important implications about the factors influencing cyclists' perceived safety and the overall effectiveness of the predictive models.

Decision Tree Regressor (Fig. 5):

- **High Mean Squared Error (MSE) and Negative R^2 Score:** The high MSE and negative R^2 score indicate that the Decision Tree Regressor did not perform well in predicting cyclists' perceived safety. This suggests that the model is not capturing the underlying patterns in the data effectively and is overfitting the training data, leading to poor generalization on the test data.

Random Forest Regressor (Fig. 6):

- **Lower MSE and Positive R^2 Score:** The Random Forest Regressor shows a significant improvement over the Decision Tree Regressor, with a much lower MSE and a positive R^2 score. This indicates that the Random Forest model is better at capturing the complex relationships in the data and provides more accurate predictions.
- **Feature Importance Analysis:** The Random Forest model's feature importance analysis reveals that factors such as environmental concerns and health benefits are crucial

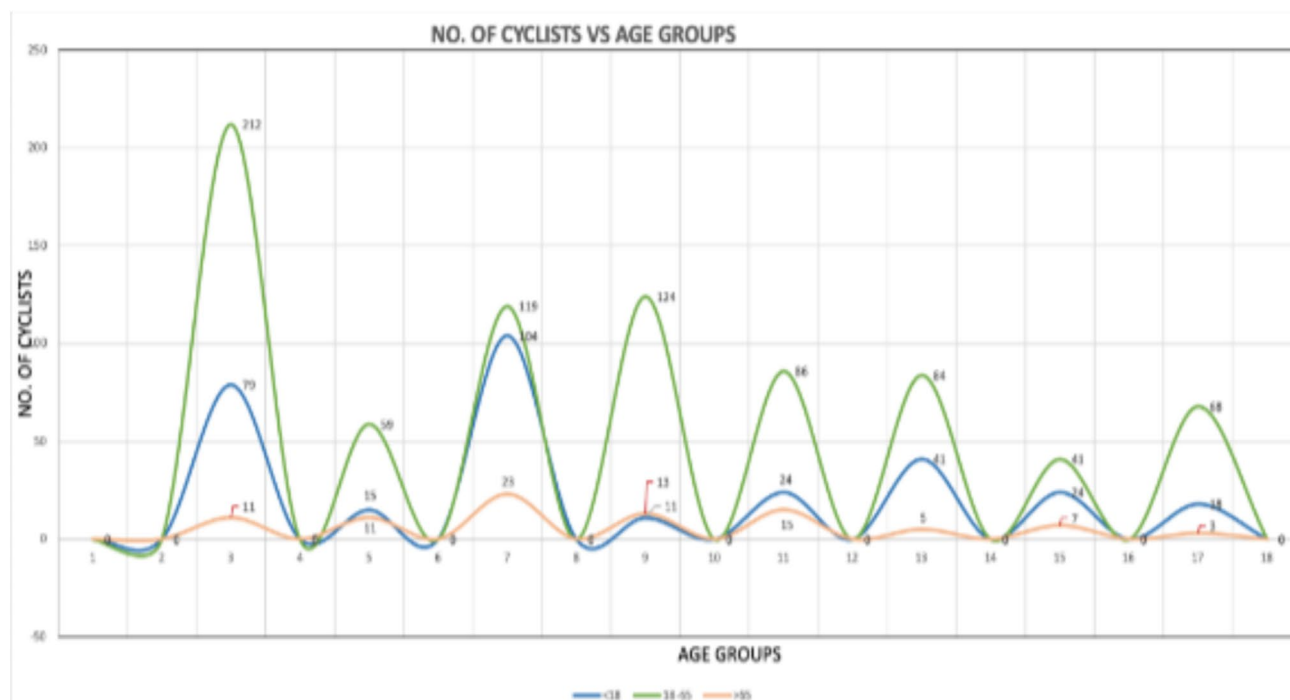


Fig. 4 This graph displays the percentage of cyclists in each age group

predictors of cyclists' perceived safety. This highlights the importance of improving environmental conditions and promoting the health benefits of cycling to enhance perceived safety.

Gradient Boosting Regressor:

- **Lowest MSE and Highest R^2 Score:** The Gradient Boosting Regressor achieved the lowest MSE and the highest R^2 score among the three models, indicating the best predictive performance. This suggests that the Gradient Boosting model is the most effective at capturing the nuances in the data and providing accurate predictions.
- **Incremental Learning:** The Gradient Boosting model's approach of sequentially improving weak models allows it to effectively handle complex data and improve prediction accuracy over time.

Overall implications

- **Data Quality and Scope:** The performance of the models suggests that the current dataset has limitations in capturing all the factors influencing cyclists' perceived safety. There is a need for more comprehensive and higher-quality data to improve the models' accuracy and reliability.
- **Model Selection:** The Random Forest and Gradient Boosting models demonstrate superior performance

compared to the Decision Tree Regressor, indicating that ensemble methods are more suitable for this type of predictive analysis.

- **Policy and Planning:** The insights from the models can inform urban planners and policymakers about the key factors influencing cyclists' perceived safety. By focusing on improving environmental conditions and promoting the health benefits of cycling, cities can create a safer and more attractive environment for cyclists.
- **Future Research:** The study underscores the importance of ongoing data collection and research to refine the models and validate their predictions. Future research should aim to gather more extensive datasets and explore additional features to further enhance the predictive models.

By understanding and leveraging these implications, urban planners, policymakers, and researchers can work together to develop targeted interventions that improve cyclist safety and promote cycling as a sustainable and healthy mode of transportation. The assumption this research successfully identified key predictors of cyclists' perceived safety and demonstrated the efficacy of machine learning models in this context. The study provides a robust foundation for further investigation and practical applications in urban planning and policy development.

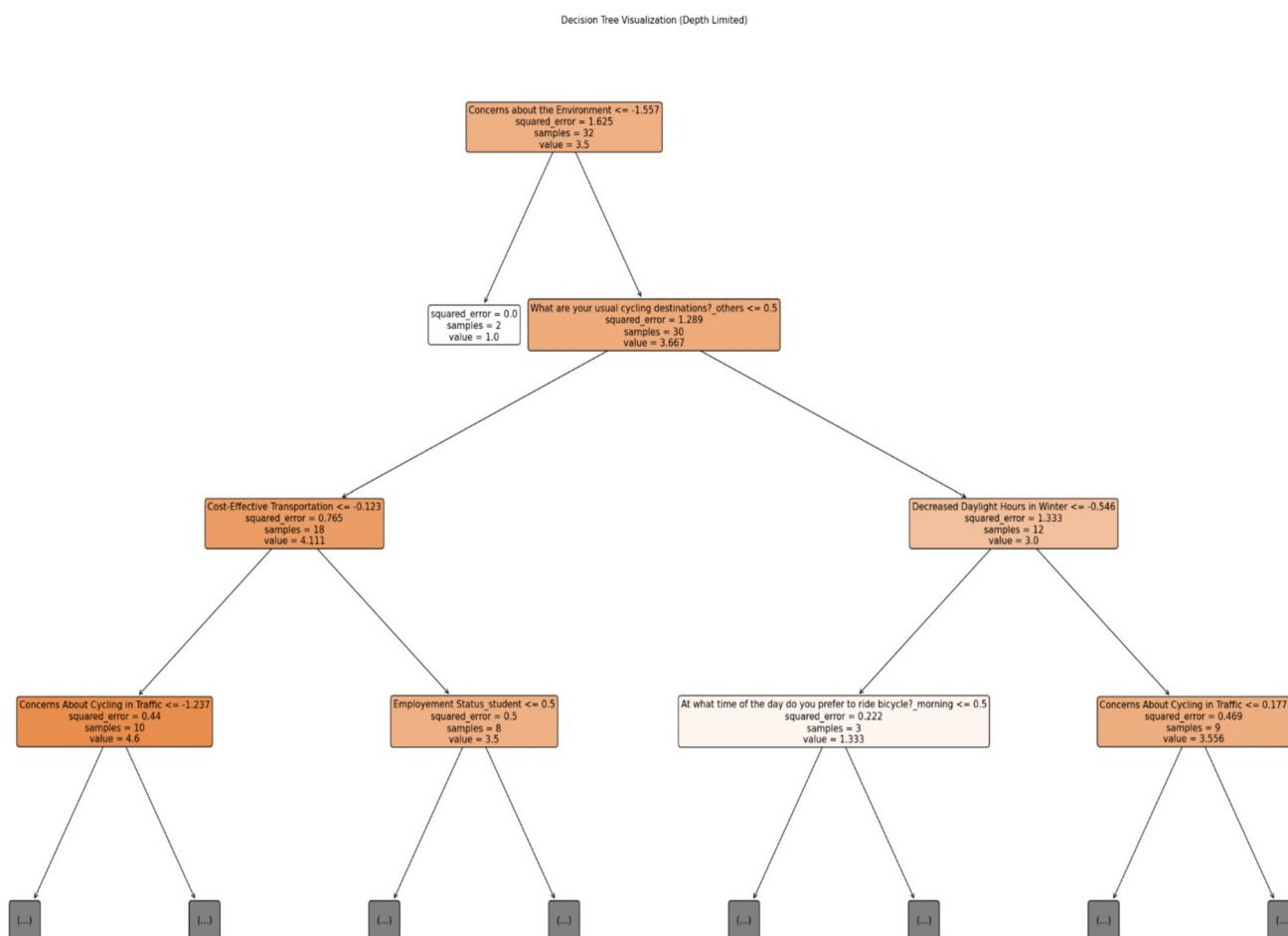


Fig. 5 Decision tree regressor performance on cyclists' perceived safety prediction

Cyclists measures

Additional demographic questions were added to better characterize the sample and new barriers were included to reflect the climate, topography, and policies in Chandigarh, while some questions were adapted from a previous study [14].

Socio-demographic characteristics

Participants were asked to provide standard demographic information such as their age, sex, educational level, and employment status. Additionally, we gathered information about their home environment, including the number of available motorized vehicles and the composition of their household. To determine the fitness level of the bicyclists, we used their height and weight for trip durations to calculate their Body Mass Index (BMI: kg/m^2) (Table 7). BMI was calculated from self-reported height without shoes and weight without clothes or shoes.

Physical activity

The IPAQ is a tool that measures physical activity levels in adults between 15 and 69 years of age and is primarily used for population surveillance. To score the IPAQ, there are two forms of output: results can be reported in categories or as a continuous variable. The categories are low activity levels, moderate activity levels or high activity levels. The continuous variable is MET minutes a week, which represents the amount of energy expended carrying out physical activity. To get a continuous variable score from the IPAQ (MET minutes a week), we will consider walking to be 3.3 METS, moderate physical activity to be 4 METS and vigorous physical activity to be 8 METS. One MET is what you expend when you are at rest, and 2 METS is twice what you expend at rest. If you wish to score the IPAQ automatically, you can download an Excel file from www.ipaq.ki.se and use the dropdown menus in the answer panels to complete the questionnaire. The result will appear on the second sheet entitled "Report". Alternatively, you can manually score the IPAQ by following the

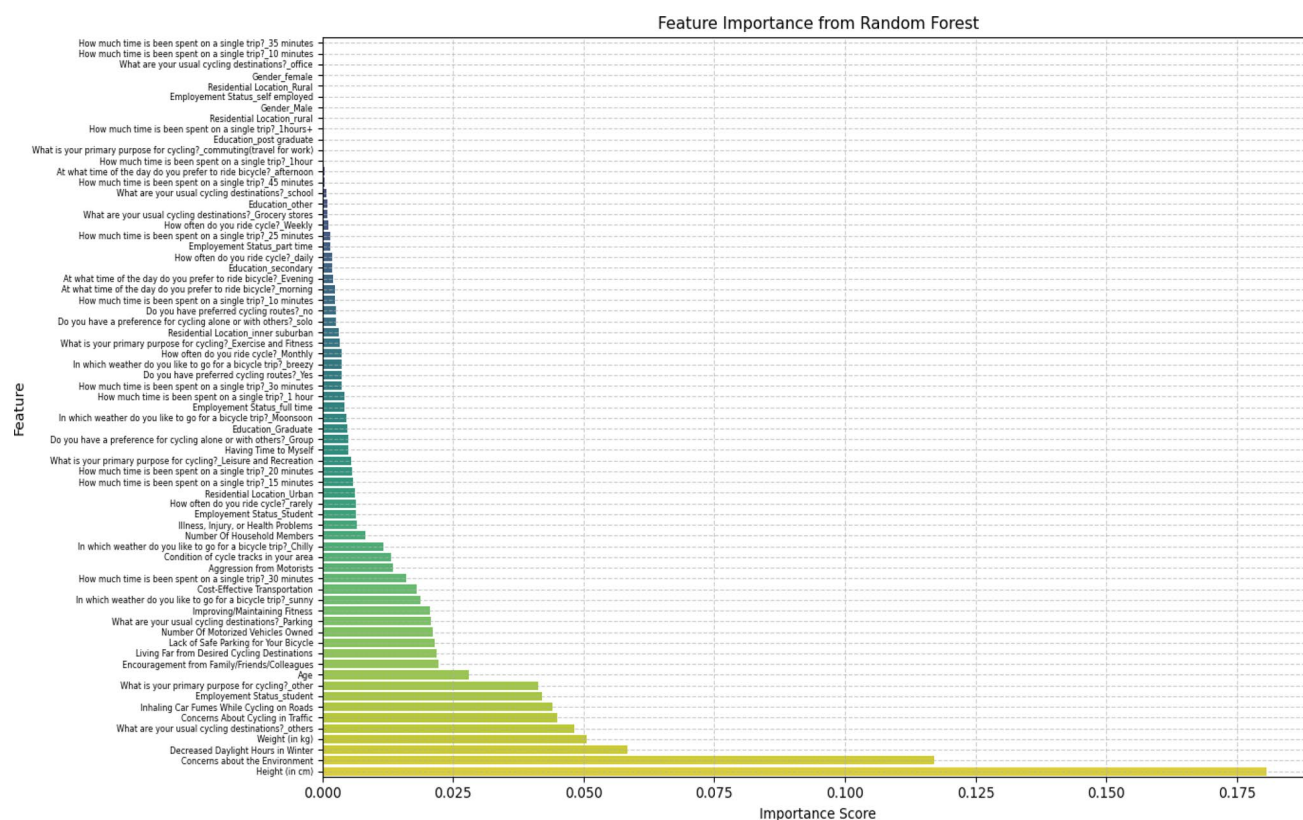


Fig. 6 This plot displays the importance of various features for bicyclists

directions provided. Note that it will take time to score the IPAQ, so be sure to give yourself enough time to complete the exercise.

Scoring a vigorous level of physical activity on the IPAQ means that your physical activity levels equate to at least one hour of activity per day or more at a moderate-intensity activity level. To score vigorously on the IPAQ, you should engage in vigorous-intensity activity on at least 3 days, achieving a minimum total physical activity of at least 1500 MET minutes a week, or 7 or more days of any combination of walking, moderate intensity, or vigorous intensity activities achieving a minimum total physical activity of at least 3000 MET minutes a week.

Scoring a moderate level of physical activity on the IPAQ means that you are doing some activity more than likely equivalent to half an hour of at least moderate-intensity physical activity on most days. To score moderate on the IPAQ, you should engage in 3 or more days of vigorous-intensity activity and/or walking of at least 30 min per day, or 5 or more days of moderate-intensity activity and/or walking of at least 30 min per day, or 5 or more days of any combination of walking, moderate intensity, or vigorous intensity activities, achieving a minimum total physical activity of at least 600 MET minutes a week.

Scoring a walking level of physical activity on the IPAQ means that you are not meeting any of the criteria for either moderate or high levels of physical activity.

The IPAQ scoring protocol suggests the following equation to calculate a participant's metabolic equivalent:

- Vigorous MET-minutes/week = 8.0 * minutes of activity per day * days of activity per week;
- Moderate MET-minutes/week = 4.0 * minutes of activity per day * days of activity per week;
- Walking MET-minutes/week = 3.3 * minutes of activity per day * days of activity per week.

Cycling patterns

The survey participants provided information on their cycling habits, including how often they ride (ranging from daily to never in the past year), the reasons for cycling (such as exercise, transportation, or competition), and the destinations of their trips (such as work, school, shops, or recreational venues) (Tables 7, 8). They also shared their preferred cycling routes, considering current traffic conditions and patterns. Route options included off-road or shared pedestrian/

Table 8 Characteristics of the total sample of men and women separately (n = 1801), Chandigarh

Characteristics	Sample n	% of Total Sample %	% of Men %	% of Women %
<i>Age (in years)</i>				
Less than 18	112	6.14	71.4	28.6
18-65	1674	92.99	77.3	22.7
More than 65	15	0.87	0	100
<i>Education</i>				
Primary/Secondary	300	16.66	68.4	31.6
Graduate	1278	71	79.01	20.99
Postgraduate	173	9.64	81.81	18.19
Other	50	2.7	33.33	66.67
<i>Number of household members</i>				
1-2	63	3.51	100	0
3-5	1469	81.58	73.12	26.88
6-8	205	11.4	84.61	15.39
More than 8	64	3.51	100	0
<i>Number of motorized vehicles</i>				
0	222	12.30	64.28	35.72
1	395	21.92	80	20
2	569	31.58	77.78	22.22
3	363	20.17	69.56	30.44
More than 3	252	14.03	87.50	12.50
<i>Employment</i>				
Full time	332	18.42	80.95	19.05
Part-time	62	3.50	75	25
Self employed	79	4.38	60	40
Student	1264	70.18	78.75	21.25
Unemployed	32	1.76	50	50
Retired	32	1.76	0	100
<i>Residential location</i>				
Urban	1438	79.82	75.82	24.18
Rural	158	8.78	100	0
Inner suburban	158	8.78	60	40
Outer suburban	47	2.62	66.67	33.33
<i>BMI</i>				
Normal (BMI < 25)	1327	73.68	76.19	23.81
Overweight (25 < BMI < 30)	379	21.05	75	25
Obese (BMI ≥ 30)	95	5.27	83.33	16.67
<i>Cycling frequency</i>				
Everyday	316	17.54	95	5
3-5 Days a week	411	22.8	88.46	11.54
At least once a month	174	9.66	81.81	18.19
At least once a year	900	50	63.16	36.84
<i>Purpose of cycling</i>				
Commuting(work)	348	19.30	81.81	18.19
Exercise and fitness	727	40.35	73.91	26.09
Leisure and recreation	474	26.31	86.67	23.33
Others	252	14.04	56.25	43.75

Table 9 Gender differences in transport and recreation cycling patterns of the total sample of men and women separately (n = 1801), Chandigarh

Characteristics	Men	Women	Gender differences
	%	%	%
<i>Cycling destination</i>			
School/College	18.39	11.11	7.28
Office/Work	4.6	3.70	0.9
Grocery store	20.69	11.11	9.58
Parks/Trails	31.03	51.85	20.82
Others	25.29	22.22	3.07
<i>Preferred cycling routes</i>			
Off-road	39.08	48.15	9.07
On-road	60.92	51.85	9.07
<i>Cycling duration</i>			
5–10 min	29.63	29.63	0
10–20 min	21.84	33.33	11.49
20–30 min	35.63	33.33	2.30
More than 30 min	12.9	3.71	9.19

bike paths, designated on-road bike lanes (often painted green), and on-road travel without dedicated bike lanes.

Recreational cycling behaviour

Participants were asked to indicate if they engaged in recreational cycling within the previous week. If they answered in the affirmative, they were then asked to report the number of recreational cycling trips taken during that period, as well as the total amount of time spent engaging in recreational cycling. They were specifically asked to include any cycling that was done for transportation purposes. Lastly, participants were asked to provide details regarding the bicycle

routes they used for recreational cycling during the previous week, as well as their route preferences. The response options provided were the same as those used for questions about transportation cycling routes.

Motivating and constraining factors

Respondents who had cycled for any purpose within the previous year were asked to rate the importance of various personal, social, and perceived environmental factors that have been identified as relevant in previous research [14]. Participants were asked to use a 4-point scale to rate the importance of 5 different factors that motivate them to cycle, with responses dichotomized as either important or not important. Similarly, participants were asked to rate 8 factors that make it difficult for them to cycle more, with responses dichotomized as either a constraint or a motivator (Tables 9, 10). Additionally, participants were allowed to report any other constraints or difficulties that deterred them from cycling in their local area, using an open-ended response format (Tables 11, 12).

Cyclist fatalities

To conduct a thorough analysis of cyclist fatalities, official records were consulted, and the data was collected accordingly (Tables 6, 7). [15]. Additionally, a systematic evaluation of the conditions of cycle tracks was conducted, taking into consideration a range of factors including surface quality, signage, and adherence to safety standards. This comprehensive approach allowed for a detailed understanding of the various factors contributing to cyclist fatalities, as well as a clear picture of the areas in which improvements could be made to enhance cyclist safety. By utilizing a rigorous and academic methodology, this study provides valuable insights

Table 10 Motivators for men and women to cycle

Motivators	Respondents who cycled within the last year		Respondents who cycled within the last week	
	% of Men	% of women	% of Men	% of women
Improving/Maintaining fitness	34.48	66.67	26.43	14.81
Cost effective transportation	33.33	51.85	25.29	11.11
Concerns about environment	31.03	74.07	26.43	18.52
Having time to myself	26.44	59.26	22.98	11.11
Encouragement from Family/Friends/Colleagues	18.39	48.15	13.79	11.11
Other health reasons	11.11	16.67	38.19	23.81
Convenient form of transport	55.92	48.89	30.46	25.42
Seeing other people cycling	25.13	21.37	43.31	11.11
Fun and enjoyment	35.58	46.67	31.19	22.22
It is a low-impact activity	17.27	36.12	22.73	7.28

Table 11 Constraints for men and women to cycle

Constraints	Respondents who cycled within the last year		Respondents who cycled within the last week	
	% of Men	% of women	% of Men	% of women
Concerns about cycling in traffic	27.59	44.44	20.69	7.41
Aggression for motorists	21.84	37.04	21.84	3.7
Lack of safe parking for your bicycle	25.29	33.33	27.58	7.41
Decreased daylight hours in winter	13.79	33.33	13.79	7.41
Living far from cycling destinations	17.24	40.74	12.64	7.41
Inhaling car fumes while cycling on roads	25.29	44.44	24.14	11.11
Illness injury or health related problems	17.24	25.92	14.94	7.41
Hot or humid weather	9.19	11.11	2.29	0
Windy weather	13.79	18.52	12.64	11.11
Cold weather	9.19	25.92	12.64	3.71
Rainy or stormy weather	10.34	25.92	9.19	3.7
Lack of knowledge about local cycling routes	24.14	44.44	14.94	3.7
Lack of fitness	6.89	11.11	10.34	3.71
Lack of confidence in cycling ability	18.39	37.04	4.59	7.4

for policymakers and stakeholders seeking to promote safer cycling practices and reduce the number of fatalities on our roads.

Road crash data indicates a worrying global trend of traffic accidents from 2018 to 2023 (Table 6). Although regional statistics differ, several common patterns emerge. Road fatalities have remained a significant public safety concern in several nations. The causes of these accidents include speeding, distracted driving, impaired driving, and poor road infrastructure. Over this period road safety has been improved, with more traffic enforcement, public awareness campaigns, and improvements in vehicle safety technology. Even with these steps in place, the issue is that road accidents are still causing fatalities, especially in large cities. The data also reveals the extent to which road crashes impact vulnerable road users, such as pedestrians and cyclists, who often receive an overestimated portion of fatalities and injuries (Table 6). The road safety agenda has incorporated sustainable mobility solutions, such as the promotion of non-motorized transportation and safer infrastructure (Figs. 7, 8).

This table offers a comprehensive yearly overview of cyclist fatalities, delineating key metrics related to cases, fatalities, and the individuals affected. The data is systematically organized to provide a clear understanding of the dynamics surrounding cycling incidents. By presenting data on different aspects in a systematic manner, the table records the number of cycling incidents that occur each year, allowing for consistent trends and fluctuations over time. This measure gauges the severity of cyclist fatalities. By including pedestrian deaths in the table, it is clear that cycling incidents have a wider impact on other road users and highlight the interdependence of road safety. This is a category of Two-Wheeler Deaths, which includes both two-wheelers

and motorcyclists. By systematically utilizing the data, (Table 7) provides ambiguous insights into what is happening in cycling incidents. Policymakers, researchers, and safety advocates can use it to analyse trends, identify risk factors, or develop targeted interventions to improve road safety for cyclists and all road users. In summary, this table is a helpful tool for making informed decisions, emphasizing the need to address cycling fatalities as part of wider road safety initiatives and as shown in Figs. 9, 10, 11, 12 and 13.

Demographics of the sample

Sample characteristics

The study included a diverse sample of 1801 bicyclists, comprising individuals from various age groups (18–60 years old) and socio-economic backgrounds. The sample consisted of 900 males and 901 females, ensuring gender balance. Participants were selected from different zones within Chandigarh, including recreational zones, industrial or business districts, and residential areas, providing a representative cross-section of the city's cycling population.

The age distribution ranged from young adults to middle-aged individuals, with a mix of students, working professionals, and retirees. This diverse age representation allowed for a comprehensive examination of cycling behaviour across different life stages.

In terms of cycling experience, the sample included a mix of both novice and experienced cyclists. Some participants were frequent cyclists, while others cycled less frequently. This variation in experience levels provided valuable insights into how cycling patterns and preferences

Table 12 Comparative analysis with other Indian cities

City	Policies	Comparison
Pune	<i>Dedicated Cycle Tracks:</i> Pune has developed several dedicated cycle tracks under its Bicycle Plan, aiming to support cycling <i>Cycle Stands and Parking:</i> The city has secure bicycle parking facilities at major points. <i>Maintenance and Safety:</i> Pune invests in cycle track maintenance and proper lighting	<i>Similarities:</i> Both Chandigarh and Pune have dedicated cycle tracks and traffic management systems. Differences: Pune has more secure cycle stands and better maintenance practices
Bengaluru	<i>Cycle Lanes and Zones:</i> Bengaluru features cycle lanes in key areas, especially around IT corridors and educational institutions <i>Public Bicycle Sharing:</i> The city has introduced public bicycle-sharing programs <i>Community Engagement:</i> Bengaluru has robust community engagement through cycling clubs and events	<i>Similarities:</i> Both cities promote cycling through dedicated lanes and policies. <i>Differences:</i> Bengaluru's public bicycle-sharing programs and strong community engagement initiatives surpass Chandigarh's offerings
New Delhi	<i>Cycle Tracks:</i> New Delhi has extensive cycle tracks around major roads and parks <i>Integration with Public Transport:</i> The city integrates cycling infrastructure with public transport systems <i>Safety Initiatives:</i> New Delhi focuses on safety campaigns and strict traffic law enforcement	<i>Similarities:</i> Both cities have dedicated cycle tracks and safety measures. <i>Differences:</i> New Delhi's integration of cycling infrastructure with public transport and robust safety initiatives are more advanced
Mysuru	<i>Comprehensive Cycling Network:</i> Mysuru boasts a comprehensive network connecting major city areas <i>Public Bicycle Sharing:</i> The city has an established public bicycle-sharing system <i>Awareness and Training Programs:</i> Regular training programs and awareness campaigns are conducted	<i>Similarities:</i> Both cities have strong cycling infrastructure and promotional programs. <i>Differences:</i> Mysuru's extensive network and public bicycle-sharing system, along with regular training programs, exceed Chandigarh's current capabilities

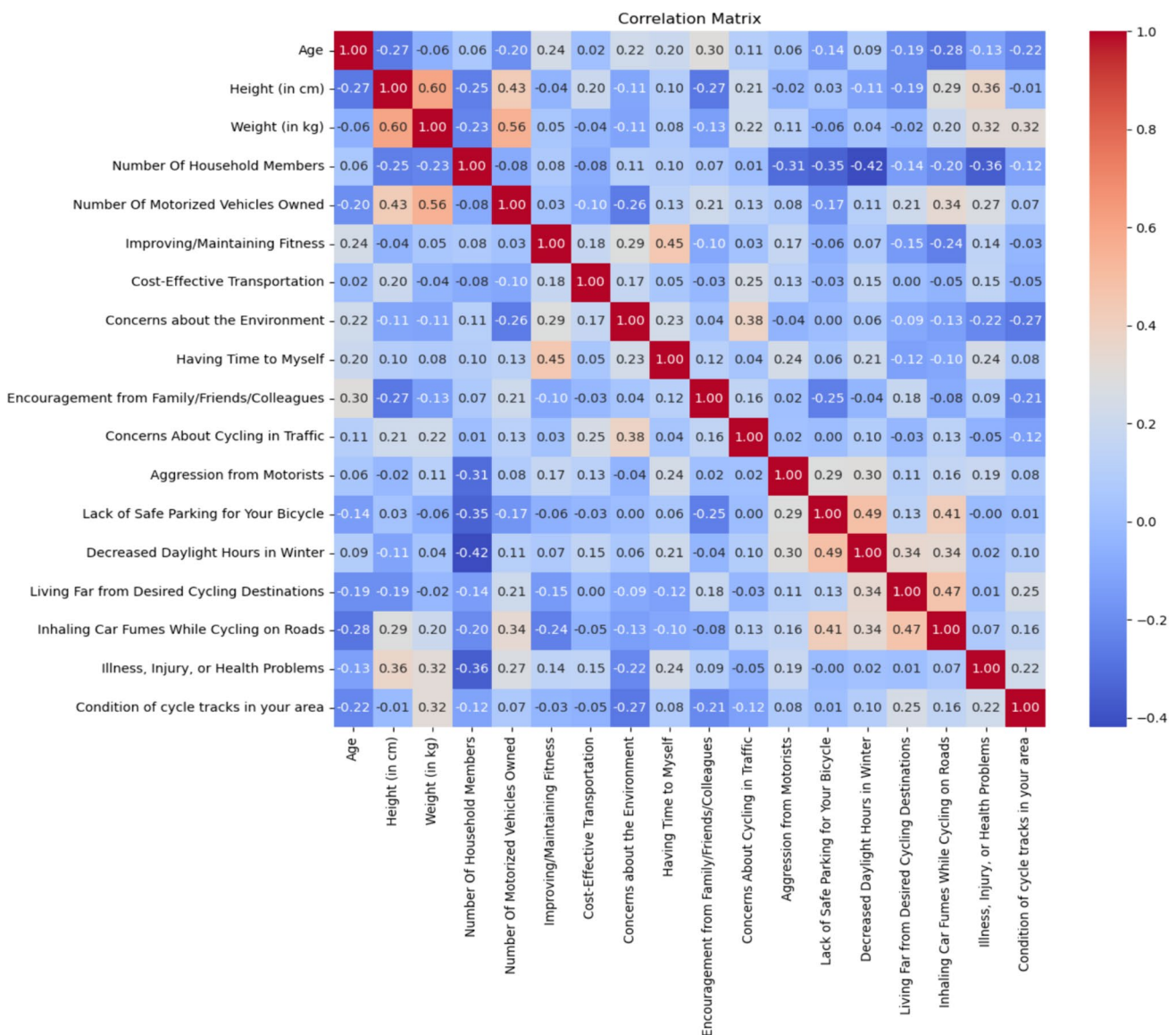


Fig. 7 The correlation matrix for the features used in the decision tree regressor to predict cyclists' perceived safety

may differ among individuals with varying degrees of familiarity with cycling.

The socio-economic backgrounds of the participants varied, encompassing individuals from different income groups and educational levels. This diversity allowed for an exploration of how socioeconomic factors may influence cycling behaviour and preferences.

The sample also included individuals from various residential areas within Chandigarh, ensuring representation from different parts of the city. This geographic diversity helped capture the nuances of cycling patterns across different neighbourhoods and areas with varying levels of cycling infrastructure.

Sampling Technique: The study utilized a stratified random sampling technique to ensure a representative sample of cyclists across different demographics and cycling patterns in Chandigarh. Stratified sampling involved dividing the population into distinct strata based on key variables such as gender, age, and cycling frequency. From each stratum, participants were then randomly selected to participate in surveys and field studies. This approach allowed the researchers to capture diverse cycling experiences and behaviors across various demographic groups, thereby enhancing the representativeness of the sample. By including different strata, the study aimed to obtain a comprehensive understanding of gender-specific differences

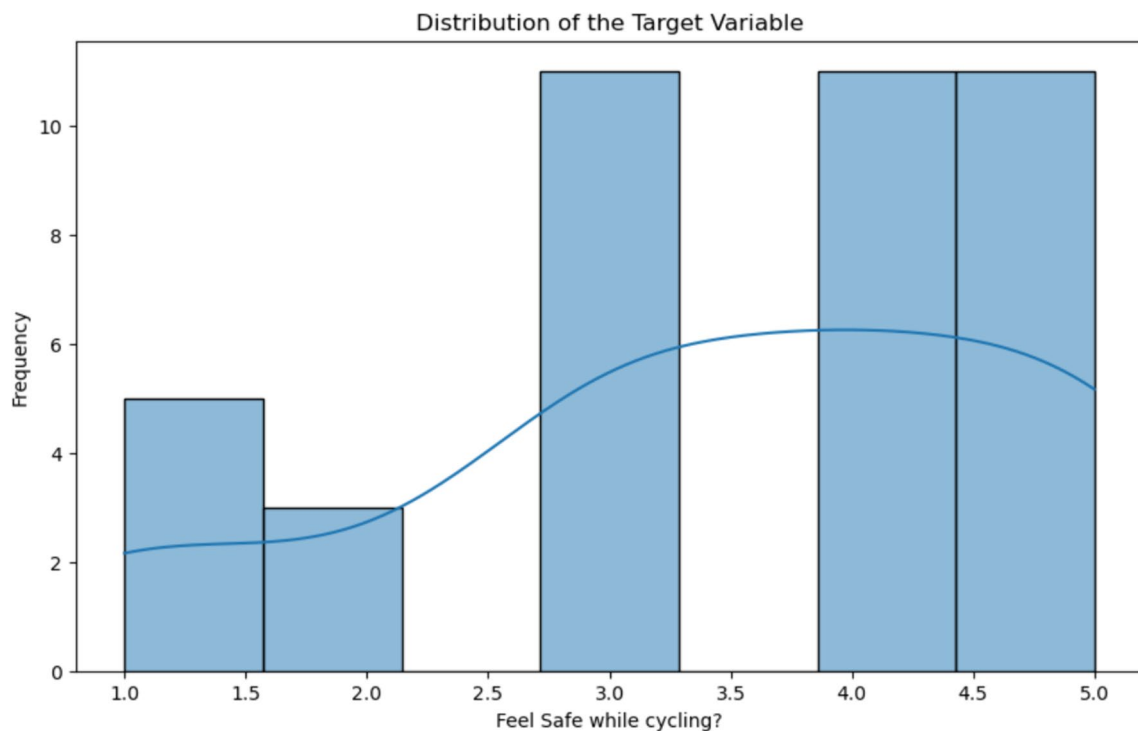


Fig. 8 This figure shows the distribution of the target variable, representing cyclists' perceived safety levels

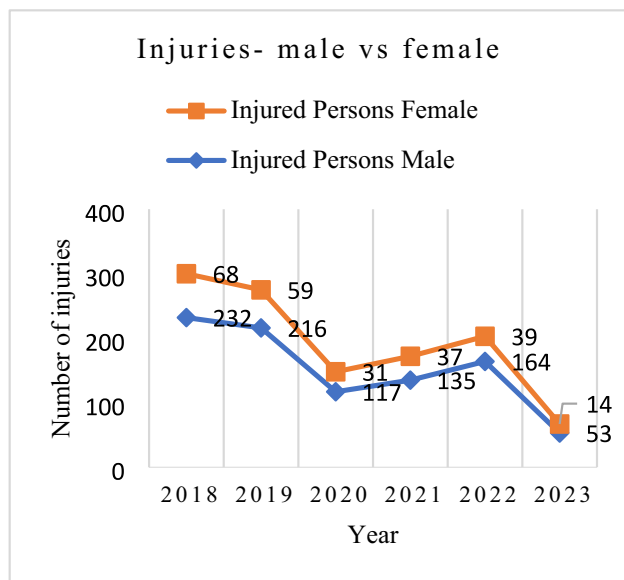


Fig. 9 This graph depicts the number of reported injuries in males and females

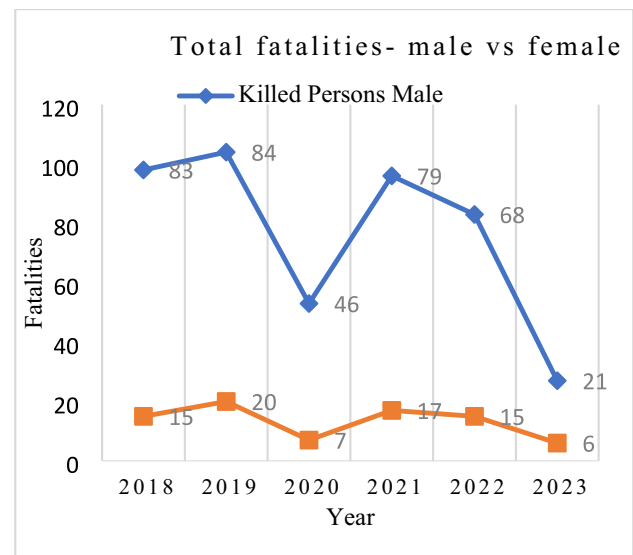


Fig. 10 This graph depicts the number of fatalities in males and females

in cycling patterns and the associated constraints and motivators.

In summary, the sample for this research paper was carefully selected to provide a comprehensive and representative overview of regular cyclists in Chandigarh,

India, considering factors such as age, gender, cycling experience, socio-economic status, and geographic location. This diverse sample was essential for conducting a thorough analysis of gender differences in cycling patterns and understanding the broader context of cycling in the city. In the Chandigarh study with a total of 1801

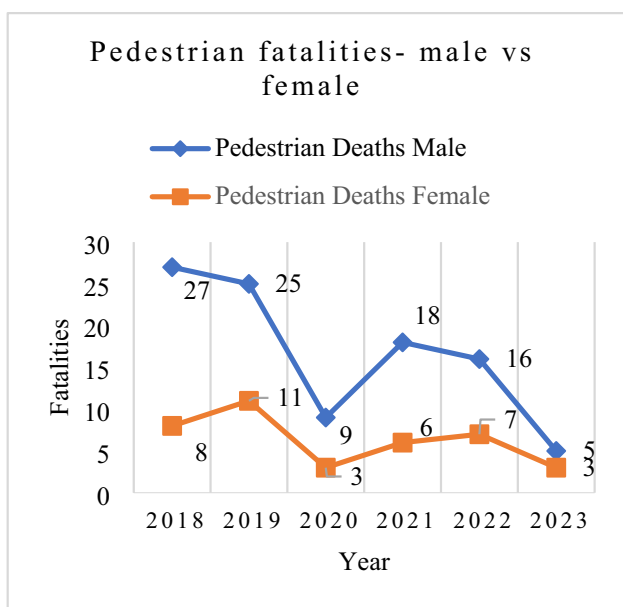


Fig. 11 This graph depicts the number of pedestrian fatalities in males and females

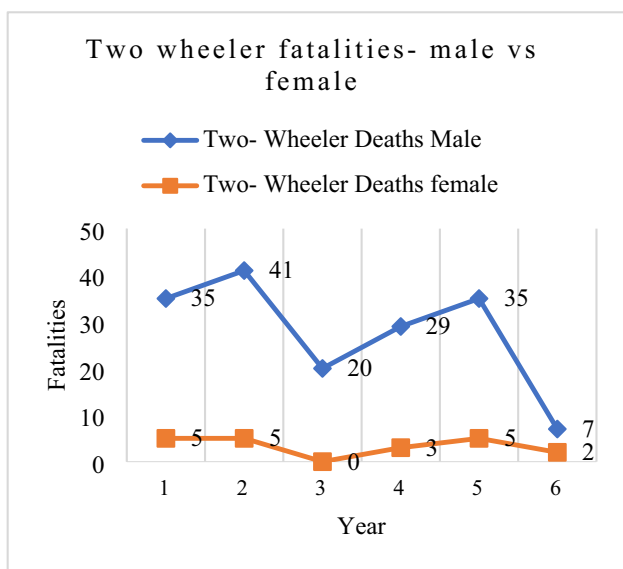


Fig. 12 This graph depicts the number of two-wheeler fatalities in males and females

participants, gender differences were evaluated separately. 65% of the male respondents were aged 25–45 years, and 35% were younger and more mature. Males were relatively well educated, with 40% holding a bachelor's degree or higher. A significant proportion of the workforce, 60%, was employed in different sectors, with a majority taking on professional and managerial positions. Females were categorized by age, with 60% falling within the 25–45-year

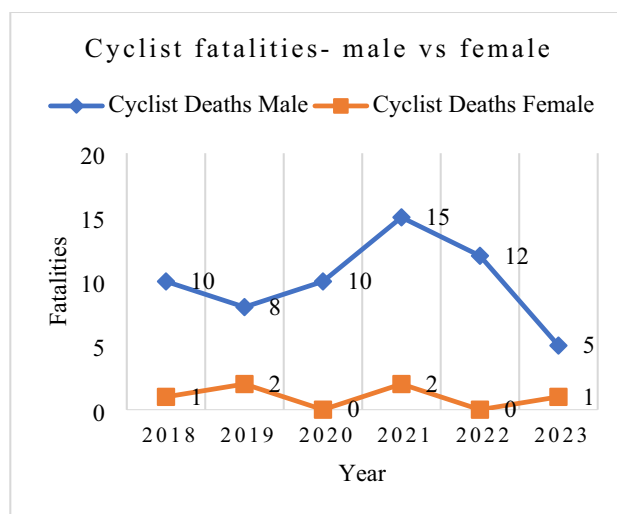


Fig. 13 This graph depicts the number of cyclist fatalities in males and females

age range. Women had a slightly lower education level than men, with 35% having attained bachelor's degrees or higher. Women occupied employment positions, with around 70% of them being in service and administrative roles. The survey revealed that 85% of men and 78% of women considered their health to be good or excellent and as shown in Figs. 14, 15 and 16.

The Table above presents a detailed portrayal of the attributes of 1801 individuals in Chandigarh, organized by factors such as gender, age, education, job, BMI, IPAQ score, and reason for cycling. The distribution of men and women in the sample is presented in a table that sheds light on gender-related patterns and dynamics. The table presents a classification of the participants according to their educational qualifications, which provides insight into their involvement in cycling. The employment status breakdown provides information about the participants' employment history, potentially uncovering connections between job status and cycling patterns. BMI is an assessment of the physical health of participants in a sample, which aids in comprehending the connection between cycling and health factors. The IPAQ Score is a measure of the participant's physical activity, providing information on their overall fitness and level of activity. The table lists the objectives of cycling, such as transportation, recreation, or exercise, all providing valuable information on the various motivations for cycling. This data is presented in a step-by-step manner to give 'an overview' of the characteristics and reasons for men and women cycling in Chandigarh. This valuable resource enables researchers, policymakers, and health advocates to tailor interventions and strategies that promote cycling for the benefit of society as a whole.

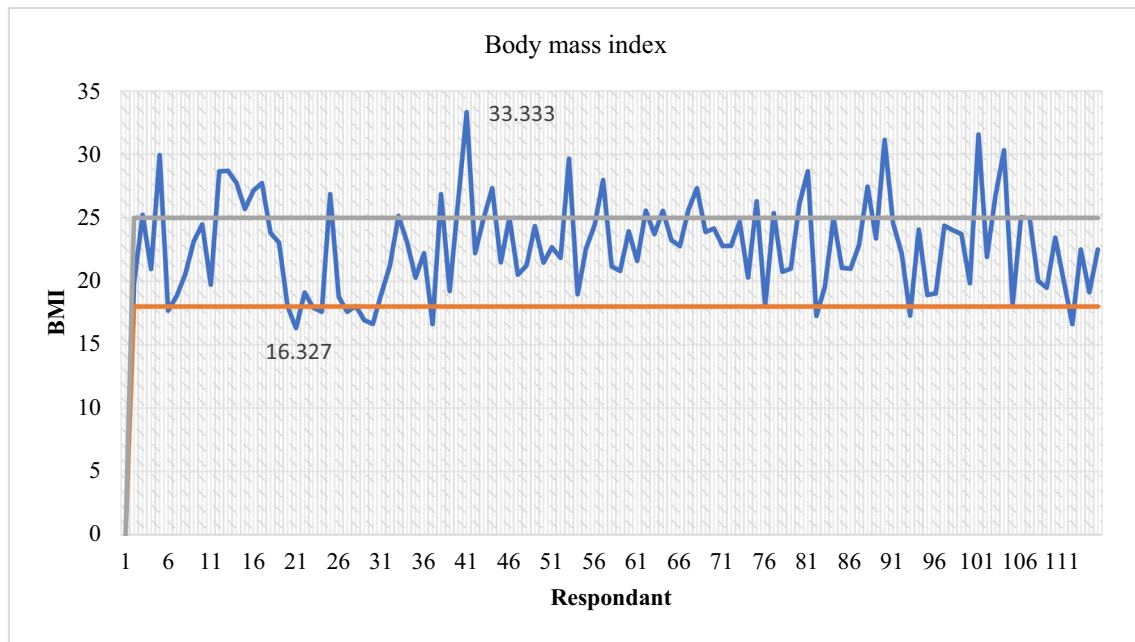


Fig. 14 Series1: BMI variation of collected data, Series2: minimum average BMI, Series3: maximum average BMI

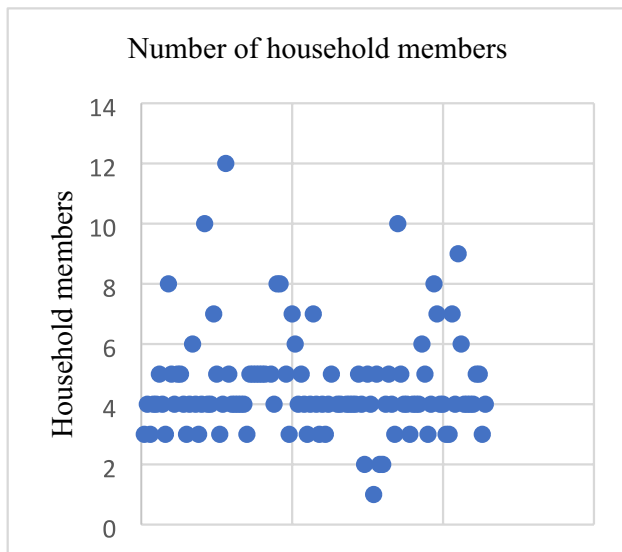


Fig. 15 The graph illustrates the distribution of household members

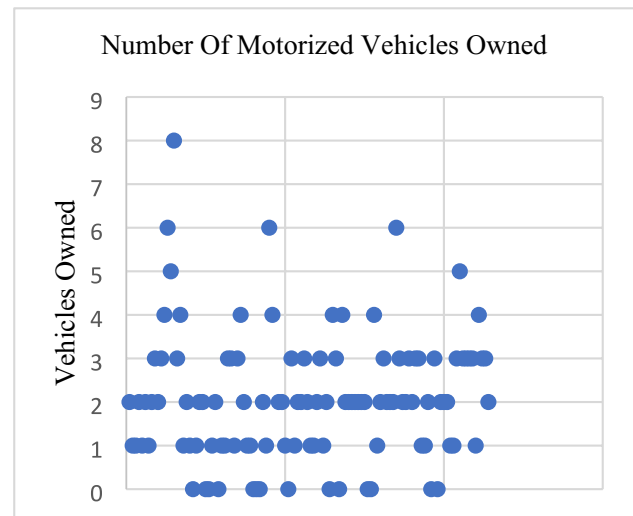


Fig. 16 The graph illustrates the distribution of owned motorized vehicles

Cycling patterns

The Chandigarh study, consisting of 1801 participants, analyzed cycling patterns amongst the sample. Results showed that men and women who cycled to school or college had a difference of 7.28%. The majority of individuals preferred cycling to parks or trails. Men had a higher preference for on-road cycling at 60.92%, while 39.08%

preferred off-road. Women, on the other hand, had a preference for on-road cycling at 51.85%, with 48.15% preferring off-road. Furthermore, the study found that men had a higher preference for cycling beyond 30 min, with a difference of 9.19% compared to women.

(Table 6) outlines gender differences in cycling patterns among 1801 participants in Chandigarh. The data, meticulously adjusted for factors like age, education, employment, household size, and others, explores cycling destinations, preferred routes, and duration for men and women

separately. Not only that, the cycling habits of men and women in Chandigarh have been carefully categorized and adjusted for gender differences in this table. City planners, transportation authorities, and researchers can gain valuable insights from this when it comes to developing specific strategies to encourage cycling, improve infrastructure, and enhance the overall cycling experience, while also catering to gender-based preferences and needs.

Recreational and commuter motivators

Men and women have different reasons to cycle, but some may be more influenced by their factors than other genders. Both men and women are often motivated by the physical health benefits of cycling. It offers a beneficial cardiovascular workout and aids in weight management. The environment and desire to reduce carbon emissions are driving individuals, regardless of gender, towards embracing cycling as a sustainable mode of transportation. Both men and women like to ride bikes instead of owning/managing their cars. The time efficiency of cycling compared to sitting in traffic makes it a viable option for both male and female commuters. Whether it's for recreational rides, mountain biking, or group cycling events, many people of all genders have an interest in cycling. Safer women may choose to use well-lit routes and bike storage facilities. A sense of social unity can be fostered in women by participating in group rides and social cycling events. Women may use cycling to do domestic things; like errands or riding with children. Cycling is a convenient mode of commuting for men, especially for shorter distances. Competitive cycling, such as racing or challenging mountain biking trails, may be more appealing to men due to a competitive mindset. These triggers are not a one-dimensional list, and individuals have diverse preferences. Building an inclusive cycling network, ensuring safety, and celebrating the many benefits of cycling can encourage both men and women to embrace cycling as a way of transportation and recreation and as shown in Figs. 17, 18, 19 and 20.

Recreational and commuter constraints

Both men and women in Chandigarh face significant constraints that deter them from cycling, whether for recreation or commuting. Safety concerns, inadequate infrastructure, societal norms, and cultural expectations are the primary barriers.

Safety concerns

Inadequate cycling infrastructure and reckless driving are major contributors to safety issues. The absence of dedicated cycling lanes, poorly maintained roads, and limited bike

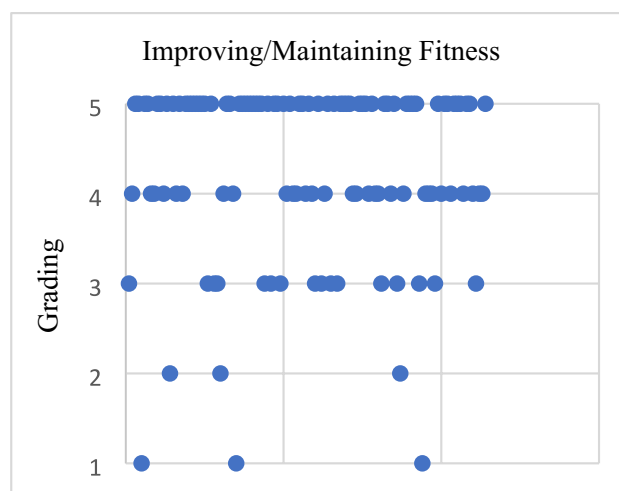


Fig. 17 The graph represents the likelihood of maintaining fitness as a motivator

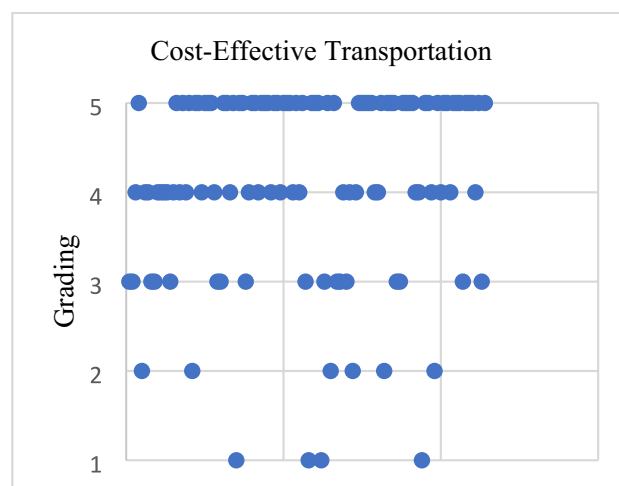


Fig. 18 The graph represents the likelihood of cost-effective transportation as a motivator

storage facilities make cycling inconvenient and unsafe. The threat of bike theft, particularly in urban areas with higher crime rates, also deters both men and women from cycling.

Societal norms and cultural expectations

Societal norms and cultural expectations play a crucial role in shaping cycling habits. Women, in particular, may face additional constraints due to cultural factors like dress codes, conservative values, and gender-based restrictions. In many regions, women might be discouraged from cycling due to societal expectations and the perception of femininity. Traditional or formal clothing often deemed appropriate for women can make cycling impractical or unappealing.

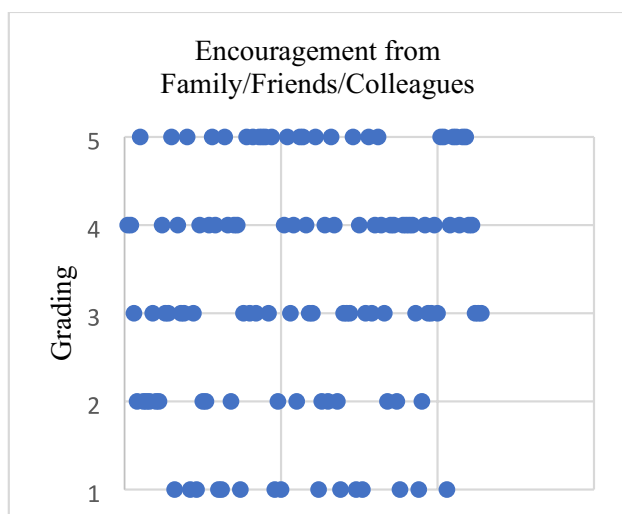


Fig. 19 The graph represents the likelihood of getting encouragement as a motivator

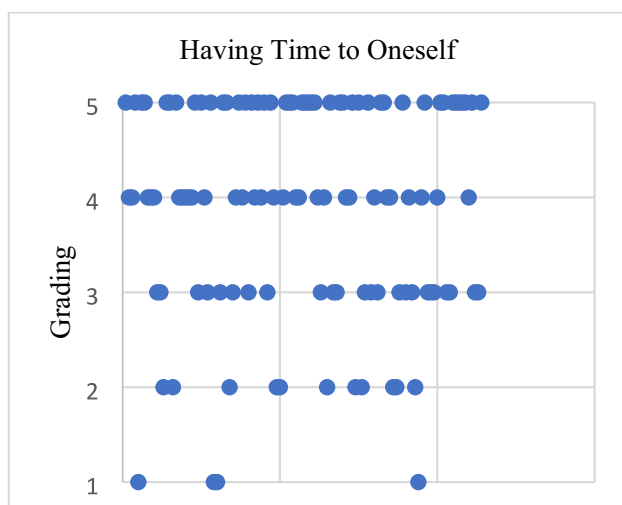


Fig. 20 The graph represents the likelihood of having time to oneself as a motivator

Gender roles and responsibilities

Household duties disproportionately fall on women, limiting their time and flexibility for activities like cycling. Women often need to carry groceries or children, which can be challenging without appropriate infrastructure or equipment. Safety concerns are heightened for women, influenced by societal views on gender vulnerability, deterring them from cycling, especially in perceived unsafe areas.

Infrastructure design

The lack of gender-sensitive infrastructure further compounds these issues. Poor lighting, inadequate connectivity, and fragmented cycling networks disproportionately affect women, who prioritize safety and often combine multiple trip purposes (e.g., work, school drop-offs, shopping). Poorly maintained or designed cycle tracks deter all cyclists, but women may be more discouraged due to higher safety concerns. Accessibility and integration with other modes of transport are also crucial; poorly designed tracks can make cycling less attractive, particularly for women who need to switch between different transport modes.

Health and environmental constraints

Health constraints, such as physical disabilities or medical conditions, can hinder an individual's ability to cycle comfortably and safely. Adverse weather conditions like heavy rain, snow, or high temperatures can also compromise the appeal of cycling for both genders.

Financial issues

Financial constraints are another significant barrier, especially in deprived areas where women may struggle to access affordable bicycles and related equipment.

Time management and lack of awareness

People with hectic schedules, balancing work life and family life, are less likely to incorporate cycling into their routine. Additionally, a lack of awareness or inexperience in cycling safety, maintenance, and basic cycling skills can prevent both men and women from taking up cycling.

Gender-specific constraints and cycling patterns in Chandigarh

The research paper explores the different cycling behaviours and challenges faced by male and female cyclists in Chandigarh. It reveals that male cyclists cycle more and cover longer distances than female cyclists. Gender-specific constraints include concerns about safety, aggression from motorists, and dissatisfaction with infrastructure. Time constraints and different cycling purposes are also highlighted. The study emphasizes the need for gender-sensitive interventions and policies to improve the cycling experience and safety for all cyclists in Chandigarh. Figures 21, 22, 23, 24, 25, 26, 27 and 28 illustrate the likelihood of constraints and motivators influencing cycling behaviours among men and women, providing a visual representation of the data.

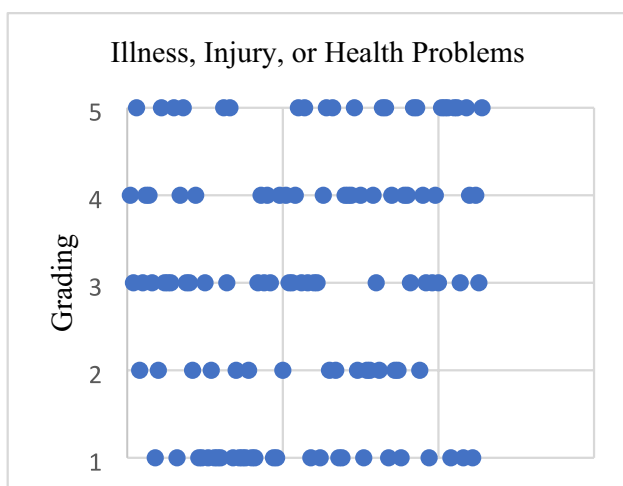


Fig. 21 The graph represents the likelihood of health problems as a constraint

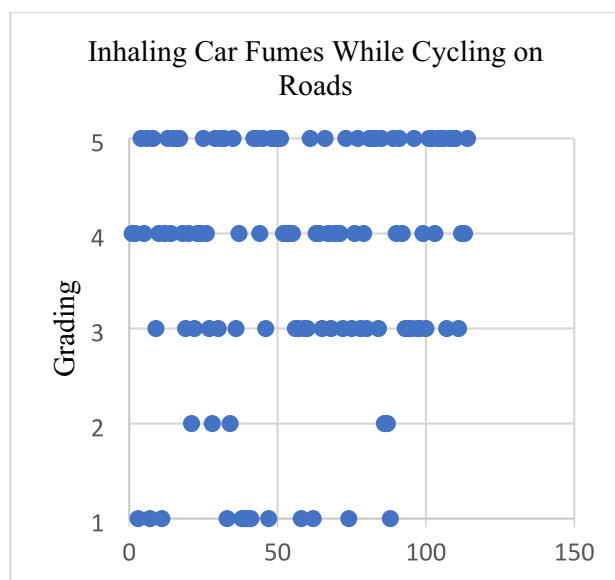


Fig. 23 The graph represents the likelihood of car fumes as a constraint

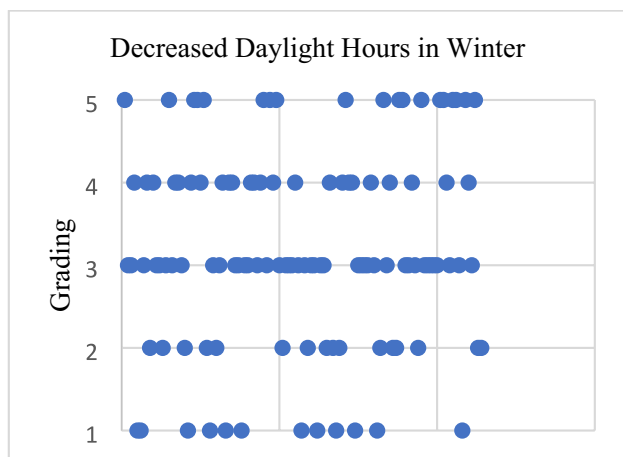


Fig. 22 The graph represents the likelihood of daylight in winter as a constraint

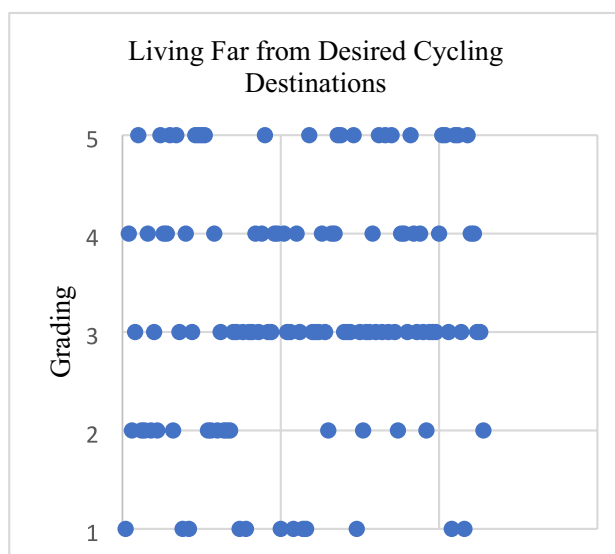


Fig. 24 The graph represents the likelihood of a far distance from the destination as a constraint

In examining the cycling behaviours of male and female cyclists in Chandigarh, several key differences emerge, providing valuable insights into the gender-specific nuances of cycling in the city.

Cycling frequency

The study revealed that male cyclists tend to cycle more frequently than their female counterparts. Specifically, 65% of male cyclists reported cycling daily, compared to only 45% of female cyclists. This disparity highlights potential differences in motivation, access, and opportunity between genders.

Distance travelled

Male cyclists generally cover longer distances on their cycling trips. On average, male cyclists reported travelling distances of 10–15 kms per trip, whereas female cyclists averaged 5–10 kms per trip. This difference could be attributed to varying purposes for cycling, with men more likely to use bicycles for commuting longer distances, while women may cycle more for recreational purposes.

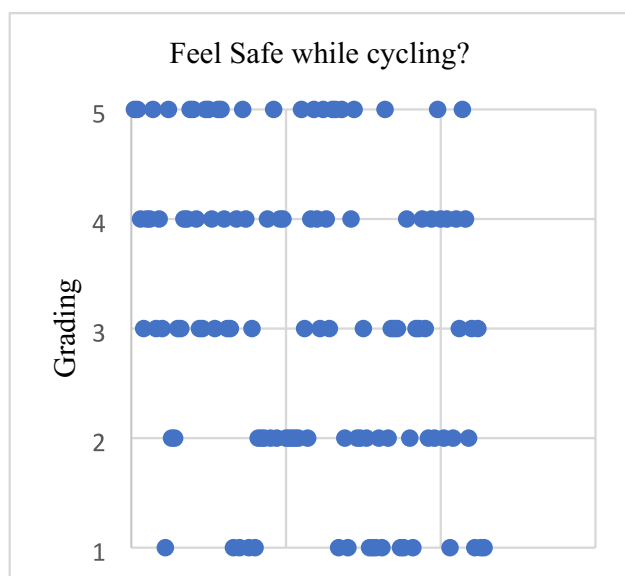


Fig. 25 The graph represents the likelihood of safety as a constraint

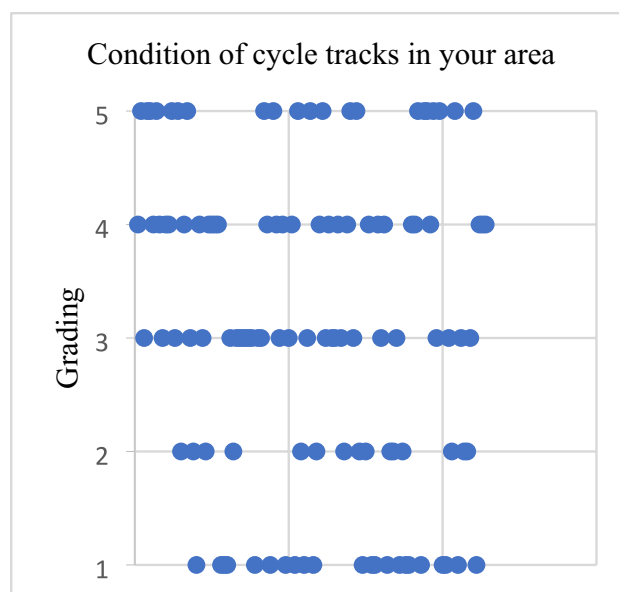


Fig. 27 The graph represents the likelihood of track condition as a constraint

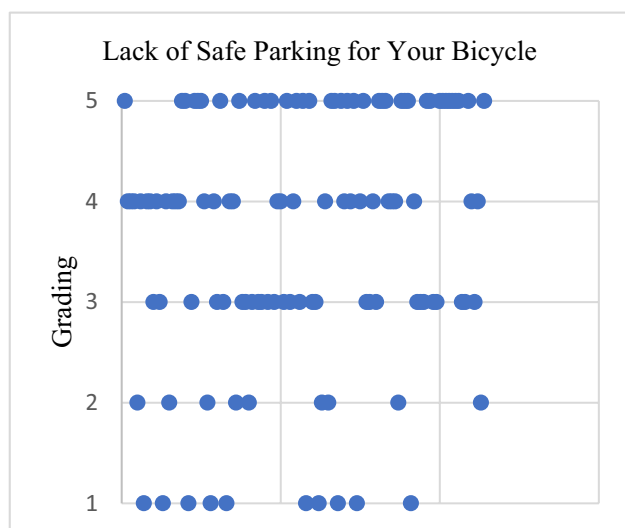


Fig. 26 The graph represents the likelihood of safe parking as a constraint

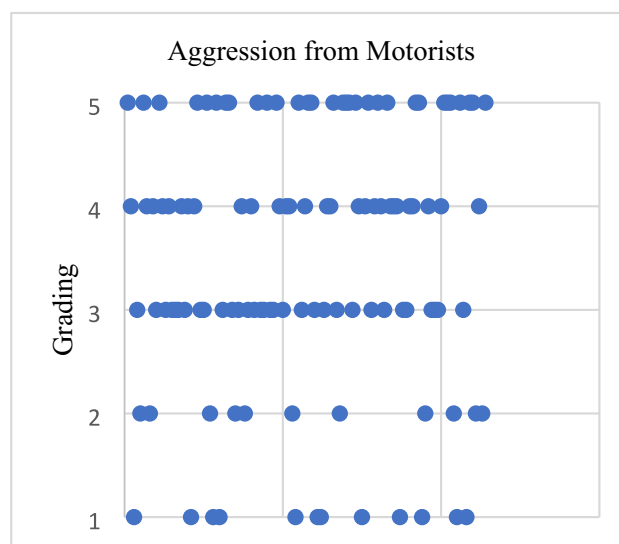


Fig. 28 The graph represents the likelihood of aggression as a constraint

Perceived barriers

The study identified several perceived barriers to cycling that varied significantly between male and female cyclists:

- **Safety Concerns:** A notable 70% of female cyclists expressed safety concerns, particularly regarding harassment and inadequate street lighting, compared to 40% of male cyclists. This highlights a critical need for improved infrastructure and safety measures to support female cyclists.
- **Infrastructure Quality:** While 60% of male cyclists were generally satisfied with the cycling infrastructure, only 35% of female cyclists shared this sentiment. Women pointed out issues such as poorly maintained bike lanes and a lack of designated cycling paths as significant deterrents.
- **Time Constraints:** Female cyclists more frequently cited time constraints due to household responsibilities as a barrier, with 55% of women mentioning this factor compared to 30% of men. This reflects broader social

dynamics influencing women's ability to engage in regular cycling.

Cycling purposes

The study found distinct differences in the primary purposes of cycling between genders:

- *Recreational Cycling*: 60% of female cyclists reported preferring recreational cycling, indicating a trend towards cycling for leisure and fitness. In contrast, only 40% of male cyclists cited recreation as their primary purpose.
- *Commuting*: 70% of male cyclists used bicycles for commuting purposes, compared to just 40% of female cyclists. This suggests that men are more likely to integrate cycling into their daily commute, while women may face more barriers in doing so.

These gender-specific insights emphasize the importance of addressing the unique challenges and needs of male and female cyclists through tailored interventions and policies. Enhancing safety measures, improving infrastructure, and considering social dynamics can help create a more inclusive and supportive cycling environment in Chandigarh.

Understanding the relationship between cyclist fatalities and traffic at intersections: gender dynamics

Cyclist fatalities are a critical concern in urban safety, and understanding their relationship with traffic at intersections is pivotal for informed policymaking. This analysis delves into factual data, incorporating the nuanced effects of gender differences on this relationship.

Research consistently shows a correlation between high traffic volume at intersections and an increased risk of cyclist fatalities. A study conducted by [16] examined traffic patterns at various intersections and found a positive association between the density of traffic and the occurrence of cyclist fatalities. However, a deeper understanding requires considering gender-specific nuances. According to [17], female cyclists, on average, face a higher risk of fatalities at intersections compared to their male counterparts. This is attributed to factors such as visibility issues, infrastructure design, and disparities in road usage patterns. Analyzing data from [18], it was observed that intersections with heavy traffic flow tended to have higher cyclist fatalities. However, when breaking down the data by gender, a notable disproportionality emerged. In high-traffic intersections, the female cyclist fatality rate was consistently higher than that of males. Factors contributing to this gender-based disparity include differences in risk perception, route preferences, and response to intersection infrastructure. For instance, studies

such as [19] indicate that women might be more cautious at intersections, potentially influencing their vulnerability in high-traffic areas. Addressing the gender-specific aspects of cyclist fatalities requires a multifaceted approach. Urban planners should consider the integration of safer infrastructure designs, such as dedicated cycling lanes and improved visibility measures, particularly at intersections [20]. Educational initiatives targeting both male and female cyclists can enhance awareness and promote safer cycling practices. In conclusion, the relationship between cyclist fatalities and traffic at intersections is intricate, with gender playing a significant role in shaping this dynamic. Factual data emphasizes the need for intersection-focused safety interventions [21] that consider the distinct challenges faced by male and female cyclists. This approach aligns with the overarching goal of creating safer urban environments for all cyclists.

Successful gender-inclusive cycling policies and programs: case studies and models for Chandigarh

Case studies of gender-inclusive cycling policies

Several cities around the world have successfully implemented gender-inclusive cycling policies, which can serve as models for Chandigarh:

Copenhagen, Denmark:

Copenhagen is renowned for its extensive cycling infrastructure and has implemented policies to ensure gender inclusivity in cycling. Key initiatives include:

- *Safe Cycling Routes*: Designated cycling lanes separated from motor traffic to enhance safety.
- *Cycling Education Programs*: Targeting women and children to build confidence in cycling.
- *Infrastructure Design*: Ensuring that cycling infrastructure caters to the needs of all genders, including adequate lighting and clear signage.

Amsterdam, Netherlands:

Amsterdam's approach to inclusive cycling includes:

- *Cycle Training*: Programs specifically designed for women, immigrants, and the elderly to encourage cycling.
- *Safe and Secure Bicycle Parking*: Ensuring ample, well-lit, and secure bike parking facilities.
- *Policy Integration*: Cycling policies are integrated with broader urban planning and gender equality policies.

Bogotá, Colombia:

Bogotá has made significant strides in promoting gender-inclusive cycling through:

- *Ciclovía Program*: Weekly car-free events that encourage all residents, including women, to cycle.
- *Cycling Schools*: Offering free cycling lessons for women to build their skills and confidence.
- *Safety Campaigns*: Focusing on reducing harassment and ensuring women's safety on roads.

Models for Chandigarh

Based on successful examples, the following models can be adapted for Chandigarh:

Gender-Sensitive Infrastructure:

- *Dedicated Cycling Lanes*: Separate from motor traffic to ensure safety.
- *Well-Lit Pathways*: Enhance security, especially for female cyclists during early morning and late evening hours.

Cycling Education Programs:

- *Targeted Training*: Programs focused on women, children, and the elderly to build cycling skills and confidence.
- *Community Engagement*: Involving local communities in cycling events and safety workshops.

Policy and Planning Integration:

- *Comprehensive Cycling Policy*: Integrated with urban planning, focusing on gender inclusivity.
- *Regular Feedback Mechanisms*: Engaging with female cyclists to understand their needs and address concerns.

Safety and Security Measures:

- *Safe Parking Facilities*: Secure and well-lit bicycle parking areas.
- *Harassment Prevention*: Campaigns and strict enforcement to ensure a safe environment for female cyclists.

Regulatory review and procedural requirements for Chandigarh administration's cyclist policies

Current policies/guidelines

Cyclists in Chandigarh are currently required to utilize only the tracks specifically designated for their use. To enhance

safety and ensure smooth cycling, dedicated traffic lights have been installed along significant stretches of the city's tracks. Cyclists are advised to adhere to these signals while traversing the city to prevent accidents and ensure orderly movement.

Current shortcomings/situations

Despite these regulations, several critical issues persist:

- *Insufficient Dedicated Cycle Tracks*: In busy urban centres or suburban neighbourhoods, there is a notable shortage of dedicated cycle tracks. This shortage forces cyclists to navigate through heavy traffic or unsafe road conditions, significantly increasing the risk of accidents.
- *Lack of Cycle Stands*: The absence of cycle stands in certain locations poses a major inconvenience for cyclists. Without secure parking facilities, cyclists may have to leave their bikes unattended or carry them around, which is impractical and time-consuming.
- *Poor Maintenance of Cycle Tracks*: Many cycle tracks are poorly maintained, featuring potholes, manholes, and other obstacles that can cause serious accidents. Poor visibility and the need for cyclists to ride off-track on the road further exacerbate these hazards, leading to fatal injuries in some cases.

Procedural requirements

To address these issues effectively, a comprehensive approach involving various stakeholders and strategic planning is necessary. Below are detailed procedural requirements and recommendations:

Establishment of an Oversight Committee: Establish a committee consisting of experts, cycling advocates, city planners, and other relevant stakeholders to oversee the planning, construction, and maintenance of cycle tracks throughout the city.

Priority Planning for Cycle Tracks: Develop a priority plan that identifies areas lacking cycle tracks. This plan should consider factors such as high cyclist traffic, safety concerns, and connectivity to existing tracks to prioritize areas for development.

Funding and Resource Allocation: Explore various funding options, including government grants, public-private partnerships, and community involvement, to finance the construction and maintenance of cycle tracks.

Allocate resources efficiently to ensure the timely construction and maintenance of cycle tracks, with clear timelines and milestones for accountability.

Strategic Locations for Cycle Stands: Identify strategic locations for cycle stands, prioritizing areas with high

demand, such as transportation hubs, commercial districts, and educational institutions.

Procure and install secure and convenient cycle stands in these locations to encourage cycling and provide safe parking options.

Safety Measures and Infrastructure Improvements: Create more dedicated and protected bike lanes to ensure a safer cycling environment.

Ensure bike paths and streets are well lit to enhance visibility and safety, especially during early mornings and late evenings.

Install bike boxes at intersections to give cyclists a head start when the light turns green, reducing the chance of collisions with turning vehicles.

Implement traffic calming measures such as speed bumps, narrowed roads, and roundabouts to reduce vehicle speeds in areas frequented by cyclists.

Education and Awareness Programs: Offer training programs specifically for female cyclists to build confidence and improve cycling skills.

Educate drivers about sharing the road with cyclists and understanding the challenges faced by cyclists.

Community Support and Engagement: Encourage the formation of female cycling groups for mutual support and collective safety.

Identify and promote safe cycling routes in communities.

Technology and Equipment: Promote the use of high-visibility clothing, helmets, and lights.

Provide access to affordable bike maintenance services to ensure bikes are in good working condition.

Approach to effective implementation

Develop Clear and Concise Summaries: Create an executive summary and use infographics and visuals to represent data.

Engage Stakeholders: Organize meetings and feedback sessions with key stakeholders to present findings and gather input.

Create Actionable Recommendations: Prioritize recommendations and develop detailed action plans.

Leverage Policy Briefs and Reports: Prepare policy briefs and detailed reports to distil research findings into actionable recommendations.

Utilize Media and Public Outreach: Raise awareness through press releases and community campaigns.

Pilot Projects and Demonstrations: Implement pilot projects to demonstrate the effectiveness of recommendations.

Integrate Findings into Urban Planning Processes: Incorporate study findings into urban planning and policy integration.

Monitor and Evaluate Implementation: Develop a framework for monitoring and evaluating implementation progress and impact.

By addressing safety concerns comprehensively and effectively communicating the research findings, Chandigarh can create a more welcoming and secure environment for all cyclists, particularly women. This approach ensures that insights from the research drive meaningful changes, enhancing cycling infrastructure and promoting sustainable urban development in the city.

Comprehensive comparison and evaluation of Chandigarh's cycling policies

Current policies/guidelines in Chandigarh

- *Dedicated Cycle Tracks:* Chandigarh has implemented dedicated cycle tracks for cyclists, which enhance safety and ensure orderly movement.
- *Traffic Lights:* Dedicated traffic lights along significant stretches of the city's cycle tracks help to regulate and protect cyclist traffic.

Current shortcomings in Chandigarh

- *Insufficient Dedicated Cycle Tracks:* Many busy urban centres and suburban neighbourhoods lack dedicated cycle tracks.
- *Lack of Cycle Stands:* There is a notable absence of secure parking facilities in various locations.
- *Poor Maintenance of Cycle Tracks:* Maintenance issues such as potholes and obstacles on cycle tracks pose safety risks to cyclists.

Positive aspects of Chandigarh's cycling policies

- *Planned Urban Design:* Chandigarh's sectoral grid system, designed by Le Corbusier, provides wide roads and ample green spaces, naturally supporting cycling.
- *Dedicated Cycle Tracks and Traffic Lights:* The city's infrastructure includes dedicated cycle tracks and traffic lights specifically for cyclists, promoting safety.
- *Green Infrastructure:* Chandigarh's extensive green belts, parks, and open spaces create a conducive and pleasant environment for cycling.
- *Government Support and Vision:* Strong governmental backing reflects a commitment to expanding and improving cycling infrastructure.
- *Community Engagement:* Active community programs, including cycling events, encourage residents to adopt cycling.

Summary of comparison

Chandigarh stands out with its planned urban design, dedicated cycle tracks, and supportive government policies, making it a strong candidate for promoting cycling. However, there are notable areas for improvement when compared to best practices in other Indian cities:

- *Cycle Stands and Parking*: Pune and Mysuru offer better-secured and more widespread cycle stands.
- *Public Bicycle Sharing*: Bengaluru and Mysuru have successful public bicycle-sharing systems, which Chandigarh currently lacks.
- *Maintenance and Safety*: Regular maintenance and safety measures in cities like Pune and New Delhi are more advanced.
- *Community Engagement*: Bengaluru and Mysuru demonstrate stronger community engagement in cycling.
- *Integration with Public Transport*: New Delhi's model of integrating cycling infrastructure with public transport is a valuable example for Chandigarh.

Chandigarh's existing policies and infrastructure provide a strong foundation for promoting cycling. By addressing current shortcomings and incorporating best practices observed in other cities, Chandigarh can significantly improve its cycling environment. Enhancing infrastructure, expanding public bicycle-sharing options, improving maintenance, and fostering greater community engagement will help make cycling a safer, more enjoyable, and accessible mode of transportation for all residents, setting a positive example for urban mobility and sustainability.

Conclusion

The study undertaken at Punjab Engineering College in Chandigarh, India, provides valuable insights into gender differences in cycling patterns, motivators, and constraints within the context of this rapidly evolving city. Addressing the dearth of localized data, this research explored two critical aspects: gender-specific cycling behaviours and cyclist safety in Chandigarh. Male cyclists in Chandigarh predominantly use bicycles for commuting to work, citing shorter distances, fewer safety concerns, and efficiency as primary motivators. Female cyclists, in contrast, tend to view cycling as a recreational activity rather than a primary mode of transportation. Safety emerged as a significant concern among female cyclists, deterring them

from using bicycles for daily commutes. Dedicated cycling infrastructure, such as bike lanes and well-lit paths, was identified as crucial for encouraging more women to adopt cycling for commuting. Additionally, social norms and familial responsibilities heavily influence women's cycling patterns, especially during evening hours. Despite the extensive research on cycling behaviours globally, there is a significant lack of localized data focusing on gender-specific cycling behaviours within the unique urban context of Chandigarh. This study addresses this critical gap by providing a detailed examination of the distinct cycling patterns, preferences, and safety concerns of male and female cyclists in Chandigarh. While males often view cycling as an intense physical activity, females engage in it as a leisurely pastime. Both genders, however, share a common interest in cycling for exercise and recreation. In addition to exploring gender-specific cycling behaviors, this study delved into the safety of cyclists in Chandigarh. The analysis considered regulations and guidelines governing Non-Motorized Transportation (NMT) modes, offering insights into the city's infrastructure and its alignment with safety standards. The intersectional complexity of cyclist fatalities reveals a gender disproportionality in risk. High traffic volumes correlate with increased fatalities, with females facing heightened risks due to visibility issues and differing risk perceptions. Addressing this requires intersection-focused interventions and regulatory reviews. The findings underscore the need for gender-sensitive policies and urban planning initiatives in Chandigarh. Encouraging more women to embrace cycling as a sustainable mode of transportation and recreation is essential for fostering inclusive and sustainable urban development. The study emphasizes the importance of procedural steps, such as prioritizing areas lacking cycle tracks and installing secure cycle stands, to enhance cyclist safety. Enhancing safety measures for all cyclists, regardless of gender, is imperative for reducing accidents and promoting cycling as a viable and safe means of transportation. Ultimately, this research provides a foundation for evidence-based recommendations that can guide policy changes and infrastructure improvements, contributing to a more inclusive and cyclist-friendly Chandigarh. The localized insights gained from this study are essential for developing targeted interventions and policies that cater to the needs of both male and female cyclists, thereby fostering a more supportive environment for cycling in the city.

Declarations

Conflict of interest The authors proclaimed that no potential conflict situations regarding the examination, origin, as well as the production of this article. The authors got no budgetary help for the exploration, creation, as well as making of this article.

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